EXCLUSION OF 61 FROM S(15): THE GENUS-8 QUOTIENT

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Result. There is no irreducible noncuspidal degree15 point on X_1(61) whose reduction at2 is the supersingular degree15 closed point. Together with the previously verified local exclusions at61, this proves 61 is not in S(15). Consequently

S(15)={2,3,5,7,11,13,17,19,23,29,31,37,43}.

The result uses exact computer assistance, as specified below.

1. THE PREVIOUS LOCAL INPUT

The complete finite-field census and local certificates in work/degree15 leave just one possible reduction for an irreducible degree15 divisor D on X=X_1(61): the unique supersingular closed point P15 of degree15. Rational-cusp reductions, all149 ordinary source orbits, all16 degree7+7 configurations with a rational cusp, and all90 degree7+8 configurations have already been excluded. There are no other supersingular or nonrational-cusp configurations of total degree15 at61. The main proof work/degree15/PROOF.txt explains these certificates and their coverage.

In particular, the present argument only has to exclude red(D)=P15. The degree15 condition is exact: a point over a degree15 field has degree 1,3,5, or15, and the first three degrees cannot support61-torsion by the known rational, cubic and quintic classifications.

Write delta_a for the geometric diamond (E,P) -> (E,aP), and put

Delta=(Z/61Z)^*/{+1,-1},
H5=<delta_3>=<delta_2^6>, H15=<delta_4>,
Y=X/H5, Z=X/H15, C=X_0(61).

The successive degrees in X -> Y -> Z -> C are5,3,2. Their genera are respectively126,26,8,4. Let s be the involution of Z induced by delta_2. Thus Z/<s>=C. All curves, maps and diamond actions are defined over Q.

2. EXACT JACOBIAN AND REDUCTION CERTIFICATES

The following properties are verified by the supplied calculations:

(i) (delta_3-1)J_X(Q) is finite;
(ii) (s-1)J_Z(Q) is finite;
(iii) J_X(Q)[2]=0;
(iv) delta_2 and delta_3 fix P15 as a closed divisor.

For (i) and (ii), finite_diamonds.m works in the sign+ cuspidal modular symbol spaces of dimensions126 and8. In projected period coordinates, the Hecke action is the transpose of DualHeckeOperator. The script checks compatibility with the ambient Hecke action through the exact projection matrix; it does not identify matrices in different bases. It recovers D2=(T2^2-T4)/2 and uses the projected winding vector e. The rows e*T2^k, 0<=k<g, span spaces of dimensions123 and7. The images of D2^6-1 and D2-1, respectively of dimensions100 and4, lie in these winding spaces. This is checked by rational matrix ranks. The winding rank-zero theorem therefore gives (i) and (ii). Inverting the diamond convention changes neither the relevant kernels nor their images.

For (iii), two61_data.m contains integral matrices on the actual sign0 Betti homology lattice of JOne(61), of rank252. For q=3,5,7, the matrices A_q=T_q-1-q*<q> annihilate rational2-torsion, by good reduction and Eichler--Shimura. Their common kernel modulo2 has dimension0. This calculation uses the actual Jacobian lattice, not a

Jacobian that is merely isogenous to it. The generating script is work/audit11/rational_two.m; the original log is work/degree15/two61.log.

For (iv), verify_reduction.m reconstructs the actual elliptic curve and61-point representing P15. Its Tate coordinates have degree15. On these coordinates delta_2, delta_3 and delta_4 act as Frobenius to the powers2,12 and4, respectively. In particular H15 acts transitively on the15 geometric points of P15, so P15 maps to a rational point on the special fibre of Z. The weaker assertion (iv) suffices below.

All these computations are exact. verify_sage.py independently checks the exported rational inclusions, common mod2 kernel, genera and the quadric calculation used in Section4. independent_checks.json records PASS. This independent linear-algebra check is not a separate agent audit.

The curve X has good reduction at2, and H5,H15 have odd order. Their tame quotients Y,Z therefore have good reduction at2, with the quotient maps and diamond actions extending over Z_2. The map f:X -> Z has degree15. Since f_* f^*=[15], its pullback is injective on2-torsion. It follows from (iii) that J_Z(Q)[2]=0 as well.

For an abelian scheme over Z_2, the torsion in the kernel of reduction on Q_2-points is killed by2: doubling carries this kernel into the modulo4 formal subgroup, on which the formal logarithm is injective. Consequently rational torsion reduction is injective on both J_X and J_Z. No assertion of injectivity on all rational points is needed.

3. DESCENT TO AN IRREDUCIBLE CUBIC DIVISOR ON Z

Suppose such a D exists. Choose a rational cusp c on X, and put
 h=delta_3. Then
 [hD-D]=(h-1)[D-15c]+15[hc-c]
is rational torsion, by (i) and Manin--Drinfeld. It reduces to zero, by (iv). Hence [hD-D]=0. A nonzero principal divisor hD-D would give a rational function over Q of degree at most15. This is impossible, since gon_Q(X)>=325*(61^2-1)/65536>15. Thus hD=D as divisors.

The order5 group H5 acts freely on characteristic-zero noncuspidal points of X: a diamond stabilizer there can only have order dividing 2 or3, from the extra elliptic automorphisms. Therefore D is the pullback of an irreducible cubic divisor D_Y on Y. Its image D_Z on Z has degree1 or3. Degree1 would give a rational noncuspidal point on X_0(61), forbidden by Mazur’s rational prime-isogeny theorem. Thus D_Z is irreducible of degree3, and f_*D=5D_Z.

Reduction commutes with f and the diamonds. Since delta_2 fixes P15, we obtain 5*(s red(D_Z)-red(D_Z))=0 in the divisor group of the special fibre of Z. That group is torsion-free, so red(D_Z) is s-invariant. (The transitive H15 action above also gives red(D_Z)=3S for one rational point S.) Choose a rational cusp c_Z of Z. Then
 [sD_Z-D_Z]=(s-1)[D_Z-3c_Z]+3[sc_Z-c_Z]
is rational torsion by (ii) and Manin--Drinfeld on the quotient. It reduces to zero, hence [sD_Z-D_Z]=0 by injectivity of torsion reduction on J_Z.

4. THE RATIONAL GONALITY OF Z IS AT LEAST4

First C=X_0(61) has no Q-map to P^1 of degree at most3. An exact smooth canonical model, returned by Magma’s SmallModularCurve(61),

is a complete intersection of degrees2 and3 in P^3. Its quadric is
 Q=x2^2-x1*x3+x1*x4+x3*x4.
Its symmetric Gram matrix has rank4 and determinant -1/4. Adjunction identifies O(1) with the canonical sheaf, so C is nonhyperelliptic. A rational degree3 pencil would give a rational ruling of this quadric: the canonical spans of the general fibres are lines contained in Q. But the nonsquare discriminant means the two rulings are exchanged over Q(i), and neither is defined over Q. This excludes a Q-map of degree3. Genus4 excludes degree1. This also agrees with the published classification: Hasegawa--Shimura, Trigonal modular curves, and Jeon, Modular curves with infinitely many cubic points, Theorem2.3. The latter is available at https://doi.org/10.1016/j.jnt.2020.09.006 . Here rational gonality is essential; C is trigonal over Q(i).

Suppose Z had a Q-map u:Z -> P^1 of degree d=2 or3. Since g(Z)=8 exceeds (d-1)^2, its degree-d pencil is unique, by the Castelnuovo--Severi inequality. For these prime degrees, two distinct pencils give a birational map to a bidegree(d,d) curve in P^1 x P^1, which would have genus at most(d-1)^2.

Thus u s=alpha u for alpha in PGL_2(Q), with alpha^2=1. If alpha=1, u factors through C. This is impossible for d=3 by parity, and for d=2 because it would give a degree1 map C -> P^1. If alpha is a nontrivial involution, pass to quotients to obtain a Q-map C -> P^1/<alpha> of degree d. The quotient of P^1 is rational over Q (for example by Luroth’s theorem), so this contradicts the preceding paragraph. Degree1 is excluded by genus. Therefore gon_Q(Z)>=4.

5. THE CONTRADICTION

The principal divisor sD_Z-D_Z has positive part of degree at most3. The gonality bound from Section4 forces it to be zero. Thus sD_Z=D_Z. The three distinct geometric points of D_Z form one Galois orbit, and s commutes with Galois. All its orbits on those three points have the same length. Since the length divides2 and must divide3, it is1. All three points are fixed by s.

Riemann--Hurwitz for the double cover Z -> C gives exactly
       (2*g(Z)-2)-2*(2*g(C)-2)=14-12=2
geometric fixed points of s in characteristic zero. Three fixed points are impossible. This excludes the last reduction at61 and proves 61 is not in S(15).

6. REPLAY AND SCOPE

Run from the workspace root, with Magma and Sage available:
```
 magma work/degree15/quotient61/verify_small_quotient.m
 magma work/degree15/quotient61/finite_diamonds.m
 magma work/degree15/quotient61/verify_reduction.m
 python work/degree15/quotient61/verify_sage.py
 python work/degree15/verify_coverage.py
```
Each Magma command should be logged to its corresponding .log file. The reduction script uses the installed mdmagma library at the absolute AttachSpec path in that script. The Sage checker also reads the prior actual-Jacobian export two61_data.m. To regenerate that export, run:
```
 magma N:=61 Qs:=3,5,7,11 Power:=6 DiamondOrder:=5 \
   MatrixFile:=work/degree15/two61_data.m work/audit11/rational_two.m
```
and save its output to work/degree15/two61.log.

The scope of the new checks is the exclusion in Sections2--5. The previous local-certificate and global-bound checks remain necessary for the full determination of S(15). The completed proof and final coverage ledger include them. Failed experiments in ss61 are historical explorations and are not used in this proof.